

HIGH PERFORMANCE SRAM DESIGN FOR ENERGY EFFICIENT SYSTEMS

Submitted in partial fulfillment of the requirements for the award of Bachelor of Engineering degree in Electronics and Communication Engineering

by

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SATHYABAMA

**INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)**

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BONAFIDE CERTIFICATE

This is to certify that this Project Report is the bonafide work of **VADDI UGANDHAR (42130516)** and **NARALASETTI UMESH SURYA KIRAN (42120311)**, who carried out the project entitled "**HIGH PERFORMANCE SRAM DESIGN FOR ENERGY EFFICIENT SYSTEMS**" under my supervision from June 2025 to August 2025.

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Dr. T. RAVI, M.E., Ph.D.,

Submitted for Viva voce Examination held on _____

Internal Examiner

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DECLARATION

We, **VADDI UGANDHAR (42130516)** and **NARALASETTI UMESH SURY KIRAN (42130311)**, hereby declare that the project report entitled “**HIGH PERFORMANCE SRAM DESIGN FOR ENERGY EFFICIENT SYSTEMS**” done by us under the guidance of **Dr. T. RAVI, M.E., Ph.D.**, is submitted in partial fulfillment of the requirements for the award of a Bachelor of Engineering degree in Electronics and Communication Engineering.

DATE:

PLACE: Chennai

SIGNATURE OF THE CANDIDATES

1.

2.

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ABSTRACT

Static Random-Access Memory (SRAM) plays a central role in modern integrated circuits, serving as the backbone for caches, embedded processors, and SoC subsystems. Conventional 6T SRAM cells, though highly area-efficient, face critical challenges when scaled to deep-submicron nodes. Issues such as degraded read stability due to direct disturbance of storage nodes, excessive standby leakage from access transistors, and poor tolerance to process-voltage-temperature (PVT) variations significantly restrict their reliable operation. These limitations become especially severe in ultra-low-voltage domains, where IoT, wearable, and biomedical applications demand both energy efficiency and robust functionality. To overcome these bottlenecks, this project investigates and enhances the 7T SRAM architecture. The seventh transistor in this design provides a dedicated read path, effectively isolating the storage nodes during read operations and thereby preserving Static Noise Margin (SNM). Building upon this baseline, two key circuit-level assist techniques are introduced. The first is **Dynamic Read Assist**, which accelerates access by strengthening bitline differentials, improving read speed and sensing margin. The second is **Adaptive Refresh**, which intelligently regulates memory refresh activity in standby mode, leading to significant reductions in leakage power without affecting data retention. The results are promising: the proposed modifications deliver up to **40% faster read access**, more than **90% reduction in standby leakage power**, and stable operation with SNM exceeding **100 mV at 0.35 V**, far outperforming conventional 6T and baseline 7T cells. These improvements establish the modified 7T SRAM as a highly energy-efficient and variation-tolerant solution. The work completed so far, including comprehensive analysis of 6T and 7T cells, lays the foundation for extending the optimized design into larger memory arrays and integrating it with decoders, sense amplifiers, and peripheral circuits.

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